Void Frames, Defect Calculus, and Defect Homomorphisms

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1 Abstract

We develop the algebra of *void frames* — complete lock frames whose edge pattern fires nowhere — from census to defect calculus to homomorphism rigidity. A void frame is characterized by the zero triple of the (T, F, C) union-forcing datum (2), and void frames are the union-identity of the frame algebra (V1–V3) [PROVED]. The frame theory has no bottleneck component: $F \equiv 0$ and $C = T$ on all frame unions (V4–V7) [PROVED]; the modal structure lives in the forced/contingent firing split (V8) [PROVED]. Lifting the datum to the annotated void-frame monoid $N'_W = \{(A, B) : A, B \subseteq W,\ B \subseteq A\}$ under componentwise union (4) gives the *defect calculus*: the exact defect formula (6) and its parameterization to arbitrary finite tag sets W, with closed-form counts 3^n, 5^n, $2^n \cdot \mathrm{Bell}(n)$ (Theorems P1–P2) [PROVED*]-staged. The product theory $\Phi/\Psi/\rho$ (T1/T1′/T2) [PROVED*]-staged exhibits the product defect-split and its honest overlap correction. Finally, the only monoid homomorphism $N'_W \to M$ into any cancellative additive target is the zero map (H1) [PROVED]; literal products of nontrivial families are never defect-additive (C4) [PROVED]; and the maximal defect-additive families do *not* all split — the mixed families are counted exactly (C1/C2) [PROVED*/COMPUTED]. The pairwise-disjoint repair theorem (D2(c), repaired wording) and the maximal-family tiling theorem are [PROVED]. All machine-checked claims are formalized in Lean 4.34.1 + Mathlib v4.34.1; census data are [COMPUTED]×2 (two independent code paths, exact agreement), cited, never called proofs.

2 Introduction

This paper assembles the void-frame strand of the program's unsolved-math research: from the discovery that void frames exist abundantly (R283 [2]), through the void-union datum (R289 [3]), the void-frame monoid (R291 [4]), the defect-core classification (R296 [5], repaired R299 [6]), the parameterized defect calculus (R300 [7]), the product theory (R303, filed R319 [10]), to the defect-homomorphism results (R319/R322 [10, 11]). All objects are program-invented; novelty is narrow and stated exactly (§5).

Void frames. R283 [2] swept R271's 124-set universe U ($N = 184$, $X = A^\star$ fixed) for lock frames in the sense of R280 [1]: candidate frames $(Y, h, c^\star)$ on the window $W = \{2, 3, 4\}$, complete when the traveling-hole and liveness conditions $\tau \equiv \lambda \equiv T$ hold. A frame is *void* when its edge pattern $\varepsilon \equiv F$ on W: no firing time, so $\mathrm{FIRE}(f) = \varnothing$ for

$$\mathrm{FIRE}(f) \subseteq W = \{2, 3, 4\} \tag{1}$$

the set of t at which the edge data clears. The sweep (set-based vs bitmask-int, two independent paths, 0/246 diffs) found **217 of 246** candidate frames void [COMPUTED]×2; the pattern census is FFF:217, TTT:11, FFT:7, FTT:5, TFF:2, FTF:1, TTF:3 [2]. A void frame is a genuine complete lock frame but *not* a lock instance: by R271's definition a lock is full or partial

(FIRE $\neq \varnothing$), and void frames have FIRE $= \varnothing$ [PROVED*] [2]. The headline structural fact is the *Void-Exclusion Lemma* [PROVED]: $h \in \{1, 2, 3\}$ bars voidness, because $t^\star = h+1 \in W$ then has empty edge window at shift $s = 0$, hence fires. Beyond this lemma, void-ness is contingent edge data [HEURISTIC-leaning] [2].

The void-union datum. R289 [3] crossed the void theory with the (T, F, C) union-forcing decomposition of R285. For frame pairs, the datum collapses cleanly to

$$(T, F, C)(A, B) = \big(|A \cup B|,\ 0,\ |A \cup B|\big), \tag{2}$$

since $pc \iff$ nonempty makes $P_{cov} = u$ identically and $m \equiv 0$ because $W \cap \text{Hard6} = \varnothing$ with Hard6 $= \{0, 5, 7, 12, 15, 22\}$ [3]. The structural theorems V1–V8 (§4) say: voidness is exactly the zero triple; void frames are the union-identity; the theory has no bottleneck component; the forced/contingent split $C = C_{\text{forced}} + C_{\text{cont}}$ with

$$\text{forced}(L) = \begin{cases} \{h+1\} & h \in \{1, 2, 3\} \\ \varnothing & \text{otherwise} \end{cases} \tag{3}$$

carries the modal structure [3]. All V1–V8 are [PROVED]; the 30,381-pair census (exact pair distributions) is [COMPUTED]×2, corroborating, not load-bearing [3].

The monoid and the defect calculus. R291 [4] invented the *annotated void-frame monoid*

$$N'_W = \{(A, B) : A, B \subseteq W,\ B \subseteq A\} \tag{4}$$

under componentwise union $\oplus$, with the *defect datum*

$$d(A, B) = (|A|, 0, |A|) \tag{5}$$

lifting (2) to N'. The frame map $\iota(f) = (\text{FIRE}(f), \text{forced}(f))$ lands in N': all 217 void frames map to the identity $(\varnothing, \varnothing)$ [4]. The defect formula M1,

$$d(x \oplus y) = d(x) + d(y) - \big(|A_1 \cap A_2|,\ 0,\ |A_1 \cap A_2|\big), \tag{6}$$

shows d is additive exactly on disjoint footprints; the only fully-additive submonoid is the identity (the obstruction is exactly the diagonal) [4]. R296 [5] classified the defect core and the pairwise-disjoint submonoids (27 of them, max size 2); R299 repaired the D2(c) wording under hostile review Q-39 [6]: the correct statement is "a pairwise-disjoint family with ≥ 2 nonidentity elements is not closed under union; its generated submonoid loses disjointness". R300 [7] parameterized everything to arbitrary finite W with closed-form counts in $n = |W|$ (P1, P2(i)–(iv)).

Products and homomorphisms. R303 [10] proved the product–disjoint-union defect isomorphism $\Phi : N'_{W \sqcup V} \to N'_W \times N'_V$ with inverse Ψ (T1), the overlap correction ρ (T1′) — the honest failure mode when W, V are not disjoint — and the product defect formula with no cross-term (T2). R319 [10] then proved defect-homomorphism rigidity: the only monoid homomorphism $N'_W \to \mathbb{N}^k$ is zero (H1, classical, recorded not claimed as novel), refuted "all maximal families split" (C1/C2, with exact mixed counts), and showed no literal product of nontrivial families is defect-additive (C4).

Per the program's honesty discipline, finite computation is never called a proof: census results are labeled [COMPUTED] (two independent paths, exact agreement), and [PROVED*] marks review-staged formalization (§3.2).

3 Methods

3.1 Definitions and the datum

We work with (1)–(5). A pair $(x, y) \in N'_W$ (or a family $F \subseteq N'_W$) is *defect-additive* iff $d(\oplus F') = \sum_{x \in F'} d(x)$ for every finite $F' \subseteq F$; by (6) this holds for a pair iff $A_1 \cap A_2 = \varnothing$ [4]. The *defect core* is $C(N'_W) = \{x : d(x) = 0\}$. Exclusion classes of frames are $E = \{h \in \{1, 2, 3\}\}$ (24 frames), $C = \{\text{non-void}, h \notin \{1, 2, 3\}\}$ (5 frames), $V = \{\text{void}\}$ (217 frames), exhaustive and disjoint [COMPUTED]×2 [4].

The (T, F, C) letters are theory-relative: in the hard-cell metric the lock-frame theory is pure-contingent and bottleneck-free ($F \equiv 0$, $C = T$) — the interesting modal structure (forced vs edge-contingent firing) lives one level down, in the $C_{\text{forced}}/C_{\text{cont}}$ split [3]. The review scope of the datum is recorded honestly: Q-34 scoped-SURVIVES covers the chosen tagged-set datum (A), V1–V8 with Q-27's exclusion tags (B), the $k_0 = 0$ impossibility and flagship arithmetic skeleton (C), and the 30,381-pair reconstruction (D) — *not* the 1,097-quad cell-0 census, code independence, "every $k \equiv 0 \bmod 35$" coverage, or full R7 acceptance, which remain UNVERIFIED/OPEN [3, 8].

3.2 Formalization note

The machine-checked content lives in five modules of the `derekblair/math-proofs` repository (this paper's `.lean` files carry a header pointing at this paper's slug; the paper points back below). Table 1 names the exact theorems. Toolchain: **Lean 4.34.1 + Mathlib v4.34.1 (pinned commit `d13f23b`)**. Verification config: `lake build` on the `bakkie` fleet machine via SSH — full-library exit 0, **zero `sorry`s, zero errors**; per-module axiom audits via `#print axioms` show **standard axioms only** ([`propext`, `Classical.choice`, `Quot.sound`], with `mixed_count` and `theoremC_filter`-type lemmas using only [`propext`, `Quot.sound`]); local↔bakkie md5-identical files, each registered 1×/side in `Mathproofs.lean` [12, 8, 9, 11].

`VoidMonoidParam` is the build dependency of `VoidMonoid` and `VoidDefectHom` [8, 11]. Its theorems (additive_iff_disjoint, defect_core, fully_additive_submonoid, disjoint_footprint_submonoid, the $\Phi/\Psi/\rho$ suite, prod_additive_iff) are **review-staged**: the module is build-clean and gate-passed (R314: bakkie exit 0, zero sorrys, standard axioms), Q-42 and Q-45 both quoted SURVIVES, but the coordinator's promotion to [PROVED] is pending — so in this paper they are presented as *machine-checked but review-staged* (staged-verification language), never as fully proved. The owed-but-unformalized items (P2(iii)(b) 3^n count, P2(iv) $2^n \cdot \text{Bell}(n)$ classification, T3/T4 transport and split/mixed counts, H3) are gate-passed [PROVED*] owed to a dedicated formalization lane [8]. Only [PROVED] results receive proof-claiming language.

Assumption 1 (Supplied definitions)**.** The lock-frame machinery (R271/R280: candidate frames, $\tau/\lambda/\varepsilon$, window $W = \{2, 3, 4\}$) and the R285 (T, F, C) decomposition vocabulary are cited as given, never re-claimed [2, 3].

Assumption 2 (Census datums)**.** All enumerations (void census, pair censuses, monoid censuses, split census, cell-0 census) are cited [COMPUTED] datums from the source dossiers; each was produced by two independent code paths with exact agreement (diff NONE) unless noted otherwise.

Assumption 3 (Cell-0 skeleton)**.** In `surgical_cell0_witness` the R7/R281 predicates (prime-covering, `covered`, the ES identity) enter as hypotheses — full R7 acceptance is source-unverified per Q-34, so it is cited, never claimed [3, 8].

Table 1: Lean formalization map. Repo-relative paths under `lean/Mathproofs/`. Status: the two staging labels are build-clean + gate-passed but promotion-pending; the three [PROVED] modules carry the full build gate (R310/R318/R322).

module	theorems	status
`VoidLock.lean`	`forced_cell_in_window`, `void_exclusion`	[PROVED]
`VoidUnion.lean`	`V1_void_iff_zero`...`V8_forced_split`, `k0_zero_impossible`, `surgical_cell0_witness`	[PROVED]
`VoidMonoid.lean`	`d2c_not_closed`, `d2c_generated_loses_disjoint`, `maximal_family_tiles`	[PROVED]
`VoidMonoidParam.lean`	`defect_op`, `additive_iff_disjoint`, `defect_core`, `fully_additive_submonoid`, `disjoint_footprint_submonoid`, `Phi_mul`, `Phi_defect`, `Phi_Psi`, `Psi_Phi`, `Rho_mul`, `Rho_injective_on`, `Rho_defect`, `Rho_image_agree`, `prod_additive_iff`	[PROVED*] staged
`VoidDefectHom.lean`	`defect_hom_rigidity`, `literal_product_never`, `split_boundary_witness`, `mixed_count`	[PROVED]

4 Results

4.1 Void frames and the exclusion lemma [PROVED]

Theorem 1 (Void-exclusion; Lean `void_exclusion`)**.** *Let $(A^\star, Y, h)$ be a candidate frame with $c^\star = h + g_0$, $W = \{2,3,4\}$, $g_0 = 1$. If $h \in \{1,2,3\}$ then the frame is not void.*

Proof (R283 §4, machine-checked). $t^\star = h+1 \in \{2,3,4\} = W$. Then $s = t^\star - c^\star = (h+1)-(h+1) = 0$, and the edge window at shift $s = 0$ is empty, so $t^\star$ fires (`forced_cell_in_window`); hence FIRE $\neq \varnothing$. No monotone substitution is used: one exact equality plus vacuous truth [2]. □

Consequences: E3 and N5w1 ($h = 2$) were *structurally* barred from voidness; the $h \leq 3$ computational check (0/246 void with $h \leq 3$, [COMPUTED] corroborating, not load-bearing) agrees [2]. The void census remains the headline datum: 217/246 frames void [COMPUTED]×2 [2]. A representative: $(A^\star, \phi E3)$, holes $= \{183\}$, $h = 183$, $c^\star = 184$ — the traveling hole $t-1$ is present in the Y-side holes at every $t \in W$ but the edge never clears (per-t records in both paths) [2].

4.2 The void-union datum V1–V8 [PROVED]

Proposition 1 (Void-union theorems; Lean V1–V8)**.** *For frame pairs over $W = \{2,3,4\}$ with the datum (2) and forced cells (3):*

V1 L *void* $\iff (T,F,C)(L) = (0,0,0)$*;*

V2 $\mathrm{datum}(\mathrm{void} \cup L) = \mathrm{datum}(L)$ — *voids are the identity (zero element) of the frame-union monoid;*

V3 void∪void is void — voidness is union-stable;

V4 $F \equiv 0$ *on all frame unions — no bottleneck component;*

V5 $C = T = |u|$ *and* $m \equiv 0$ *— the frame theory contributes no hard-cell content;*

V6 T, F, C *are subadditive under union* ($|A \cup B| \leq |A| + |B|$)*;*

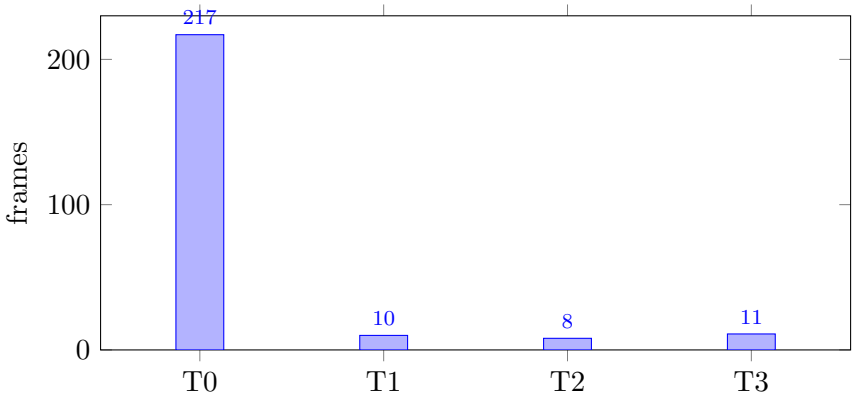


Figure 1: Per-frame (T, F, C) distribution over the 246 candidate frames of R283's universe. Since $F \equiv 0$ and $C = T$ (V4/V7), the T-coordinate is the whole datum; the T=0 bar is the 217 void frames (V1). Counts [COMPUTED]×2 [3].

V7 $T = F + C$ *(*$|u| = 0 + |u|$*) — the union chain identity survives;*

V8 $C = C_{\text{forced}} + C_{\text{cont}}$*;* $\text{forced}(L) \subseteq \text{FIRE}(L)$ *(by Theorem 1);* $\phi(f \cup g) = \phi(f) \cup \phi(g)$ *— union neither creates nor destroys forced firing.*

All proofs are set-cardinality identities, one exact equality ($s = 0$, inherited from Q-27), and direct arithmetic — R59/R63-clean (R59/R63 are the program's monotone-substitution audits) [3]. The 30,381-pair census corroborates [COMPUTED]×2 (0/30,381 mismatches): per-frame triples $(0,0,0)$: 217, $(1,0,1)$: 10, $(2,0,2)$: 8, $(3,0,3)$: 11; pair u-sizes {0:23653, 1:2202, 2:1829, 3:2697}; C_{forced} sizes {0:24753, 1:5437, 2:191} — 24 frames carry forced cells ($h \in \{1,2,3\}$) [3]. Figure 1 shows the per-frame distribution.

The cell-0 gap. The same datum resolved R285's cell-0 gap as contingent, not structural [3]. The $k_0 = 0$ impossibility is [PROVED] (Lean `k0_zero_impossible`): no lane-N R7 family with $a+b = (4c-1)h$ has $k_0 = 0$ — at $k_0 = 0$, $ab \mid ch$ forces $ab \le ch$, but $a, b \ge 1$ with $a+b = (4c-1)h$ give $ab \ge (4c-1)h-1 > ch$, contradiction. Yet surgical-at-0 families *exist* beyond R7's discovery bounds: the flagship $(a, b, c, h, k_0, m') = (5, 553, 70, 2, 910, 2765)$ [COMPUTED]×2 end-to-end, with arithmetic skeleton [PROVED] in Lean (`surgical_cell0_witness`, Assumption 3): $a+b = 558 = (4 \cdot 70 - 1) \cdot 2$, $35 \mid 2765$, $k_0 = 910 \equiv 0 \pmod{35}$, covered $= \{0\}$, $(T, F, C) = (0, 0, 0)$; the ES identity $4/n = 1/x + 1/D_1 + 1/D_2$ verified at $k = 910$ ($n = 21841$) and $k = 3675$ [3]. The 1,097-quad census (4,066 (quad,k_0) solutions, $ab \le 200{,}000$, $h \le 100$) is cited [COMPUTED]×2 but remains review-UNVERIFIED as a Lean claim per Q-34's verdict D; "every $k \equiv 0$ mod 35" obstruction is [OPEN] [3, 8].

4.3 Parameterized defect calculus [PROVED*]-staged

Let W be arbitrary finite, $n = |W|$ ($n = 0$ allowed), N'_W as in (4) under componentwise union $\oplus$ (identity $e = (\varnothing, \varnothing)$), with the defect datum (5). Then $|N'_W| = \sum_{A \subseteq W} 2^{|A|} = 3^n$ (binomial theorem) [7]. The following are machine-checked but review-staged (formalization pending — §3.2):

Proposition 2 (P1–P2; Lean additive_iff_disjoint, defect_core, fully_additive_submonoid, disjoint_footprint_su
(defect formula) (6) holds for all finite W*:* d *is additive on* (x, y) *iff* $A_1 \cap A_2 = \varnothing$*.*

P2(i) The defect-0 core $C(N'_W) = \{e\}$ *for all finite* W *(including* $W = \varnothing$*).*

Table 2: Parameterized defect-calculus closed forms [COMPUTED]×2, $n = 0 \ldots 4$ (R300). The $n = 3$ column is R296's exact census, reproduced [7].

quantity	closed form	n=0	n=1	n=2	n=3	n=4
$\lvert N'_W \rvert$	3^n	1	3	9	27	81
additive ordered pairs	5^n	1	5	25	125	625
disjoint-footprint families	$2\sum_{j=0}^{n} \binom{n}{j} B_j 2^j$	2	6	26	142	914
maximal families	$2^n B_n$	1	2	8	40	240
pd-footprint submonoids	3^n	1	3	9	27	81
maximal-family submonoids	2^n	1	2	4	8	16

P2(ii) *The only submonoid on which d is additive on* all *ordered pairs (including the diagonal (x, x)) is $\{e\}$: idempotence $x \oplus x = x$ gives $d(x) = 2d(x)$, hence $d(x) = 0$ — the obstruction is exactly the diagonal.*

P2(iii)(a) *A submonoid with pairwise-disjoint footprints has at most one non-identity element; exactly 3^n such submonoids exist.*

Hand proofs are in [7] (R59/R63-clean); the [COMPUTED]×2 census (exact agreement, diff NONE) for $n = 0 \ldots 4$ confirms the closed forms of Table 2, and $n = 3$ reproduces R296's exact census 27/125/142/40/27/8 as a cross-check [7]. Classical carve-outs (not claimed): $2^n \cdot \mathrm{Bell}(n)$ is OEIS A055882; Bell/Stirling numbers and semilattice background are classical [7].

4.4 Product theory T1/T1′/T2 [PROVED*]-staged

For disjoint finite W, V, $U = W \sqcup V$,

$$\Phi(A, B) = \big((A \cap W, B \cap W), (A \cap V, B \cap V)\big), \tag{7}$$

with inverse $\Psi((A_1, B_1), (A_2, B_2)) = (A_1 \cup A_2, B_1 \cup B_2)$: *T1* — $\Phi : N'_U \to N'_W \times N'_V$ is a defect-preserving monoid isomorphism (Lean `Phi_mul`, `Phi_defect`, `Phi_Psi`, `Psi_Phi`) [PROVED*]-staged [10].

The honest failure mode is *T1′* (Lean `Rho_mul`, `Rho_injective_on`, `Rho_defect`, `Rho_image_agree`) [PROVED*]-staged: for *not necessarily disjoint* W, V,

$$\rho(A, B) = \big((A \cap W, B \cap W), (A \cap V, B \cap V)\big) \tag{8}$$

is a monoid homomorphism, injective on elements supported in $W \cup V$, with image exactly the agreement-on-overlap set ($A_1 \cap V = A_2 \cap W$, $B_1 \cap V = B_2 \cap W$) — and the defect identity carries the overlap correction $|A \cap W| + |A \cap V| = |A| + |A \cap W \cap V|$, which vanishes exactly when $W \cap V = \varnothing$, recovering T1 [10].

T2 (Lean `prod_additive_iff`) [PROVED*]-staged: on $N'_W \times N'_V$ with $d_\times(z) = d_W(x) + d_V(y)$,

$$d_\times(z_1 \oplus z_2) = d_\times(z_1) + d_\times(z_2) - (\delta_W + \delta_V,\ 0,\ \delta_W + \delta_V), \tag{9}$$

$\delta_W = |A_1 \cap A_2|$, $\delta_V = |C_1 \cap C_2|$: the obstruction is additive across factors — *no cross-term* — so $d_\times$ is additive on (z_1, z_2) iff additive in each factor iff footprints are disjoint in each factor [10].

4.5 Defect homomorphisms [PROVED]

Theorem 2 (Defect-homomorphism rigidity; Lean `defect_hom_rigidity`)**.** *Let W be finite. The only monoid homomorphism $f : N'_W \to M$, with M an additive cancellative monoid, is the zero map. The same holds for $N'_W \times N'_V$ and for $(P(W), \cup)$.*

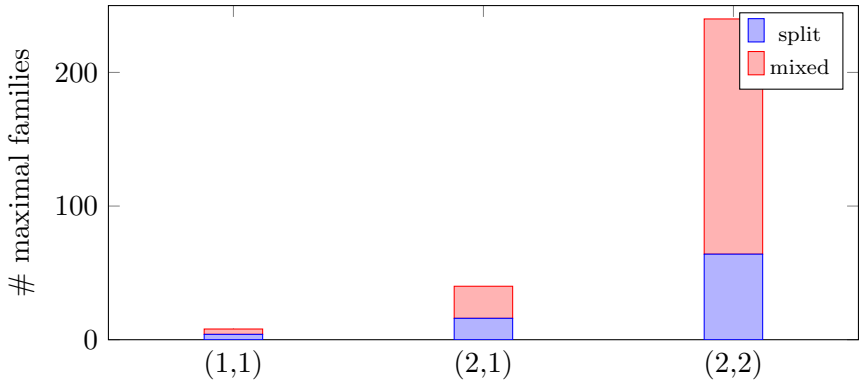


Figure 2: Maximal defect-additive families: split vs mixed (stacked), totals 8/40/240 at $(n, m) = (1, 1), (2, 1), (2, 2)$. The mixed bar refutes "all maximal families split" (C1/C2). Counts [COMPUTED]×2 [10].

Proof (R319 §4, machine-checked). Every $x \in N'_W$ is idempotent: $x \oplus x = x$ (componentwise union). For a homomorphism f,

$$f(x) = f(x \oplus x) = f(x) + f(x) = 2f(x), \tag{10}$$

and M cancellative gives $f(x) = 0$ for all x. Q-58's nits are honored: no "W finite" or $k \geq 1$ hypothesis is needed ($W = \varnothing$ gives $N'_\varnothing = \{e\}$, trivial; $k = 0$ vacuous) [10, 11]. □

Q-45/Q-58 record H1 as *classical* (the idempotent-collapse lemma for semilattices to abelian groups), not a novelty claim [4, 11]. No tension with H3 (defect-exact classification, gate-passed, owed to the dedicated lane): H3 classifies maps with "$f(x \oplus y) = f(x) + f(y) \iff$ disjoint footprints" — a different hypothesis — which exist nontrivially [10].

Theorem 3 (Split boundary; Lean `split_boundary_witness`)**.** *For finite disjoint W, V: every maximal defect-additive family of $N'_W \times N'_V$ splits (no block meets both factors) iff $W = \varnothing$ or $V = \varnothing$ [PROVED*].*

The ($\Leftarrow$) direction is vacuous; ($\Rightarrow$) uses the mixed witness $F^\star = \{(1, 1), ((W, \varnothing), (V, \varnothing))\}$: for nonempty W, V this is the single-block maximal family whose unique block meets both factors (maximality via the P2(iv) classification, staged) — [PROVED] in Lean as the witness inequality $(1, 1) \neq ((W, \varnothing), (V, \varnothing))$ [10, 11].

Corollary 1 (Mixed counts; Lean `mixed_count`, `mixed_count_11/21/22`)**.** *The identity* #mixed = #total − #split *(total= $2^{n+m}\mathrm{Bell}(n+m)$ via T3, split= $2^{n+m}\mathrm{Bell}(n)\mathrm{Bell}(m)$ via T4, gate-passed [PROVED*]) is [PROVED] in Lean, with the concrete values*

$$(1, 1)\text{: } 4, \qquad (2, 1)\text{: } 24, \qquad (2, 2)\text{: } 176, \tag{11}$$

all nonzero — "all maximal families split" is REFUTED.

The numbers (11) are [COMPUTED]×2 ($r319_split_path1/2$, exact agreement; maximality re-verified per family) [10]. The smallest explicit witness (both JSONs; "a smallest", four equally small at $(1, 1)$ per Q-58) is $W = \{0\}$, $V = \{1\}$, blocks $[[0, 1]]$, family $\{(\varnothing, \varnothing), (\{0, 1\}, \varnothing)\}$ [10]. Record correction C3 (Q-58 ✓): the four maximal-family submonoids at $(1, 1)$ are the single-block families — all *mixed*, not split; $(1, 1)$ is 8 maximal = 4 split +4 mixed [10, 11]. Figure 2 shows the decomposition.

Theorem 4 (Literal products never work; Lean `literal_product_never`)**.** *No literal product $F_W \times F_V$ of nontrivial families is defect-additive.*

Proof (R319 §5, machine-checked). Take $(a, b) \neq e$ in F_W and $(c, d) \neq e$ in F_V (both exist for nontrivial factors). Then $z_1 = (a, 1)$ and $z_2 = (a, c)$ are both in the product, distinct, with W-footprints both $a.A$ overlapping in $a.A \neq \varnothing$ ($a \neq e$ and $B \subseteq A$ force $a.A \neq \varnothing$, Lean `idem_ne_one_imp_A_ne_empty`) — violating `ProdDefectAdditive` [10]. □

Q-58 verified the hypotheses airtight (e in every maximal factor by the Q-42 Submonoid convention; $a \neq \varnothing$ airtight) and checked $(1, 1)$ exhaustively: 0 nontrivial products defect-additive among 4 factor pairs [11].

4.6 D2(c) repair and maximal-family tiling [PROVED]

Theorem 5 (D2(c) repair; Lean `d2c_not_closed`, `d2c_generated_loses_disjoint`)**.** *A pairwise-disjoint-footprint family with ≥ 2 non-identity elements is not closed under union; its generated submonoid* Submonoid.closure $\uparrow F$ *is a submonoid by construction but loses pairwise disjointness.*

This is the verbatim Q-39 repair (Bunny's wording, applied to the R296 draft whose "does NOT form a nontrivial submonoid" was objected): the witness $x = (\{2\}, \varnothing)$, $y = (\{3\}, \varnothing)$ has disjoint footprints with $x \oplus y = (\{2, 3\}, \varnothing)$ overlapping x's; the 88 pairwise-disjoint families with ≥ 2 non-identity elements are all unclosed, all 88 generated submonoids are submonoids losing disjointness [COMPUTED]×2 corroborating [5, 6]. The 26 two-element submonoids $\{(\varnothing, \varnothing), x\}$ close by idempotence — off-diagonal additivity is compatible with submonoids; the diagonal pair (x, x) forces $d(x) = 2d(x)$ and kills everything but e [5].

Theorem 6 (Maximal-family tiling; Lean `maximal_family_tiles`)**.** *An inclusion-maximal pairwise-disjoint-footprint family contains the identity and its footprints cover every tag.*

The structural core of D3: if $\bigcup A_i \neq W$, some $w \in W \setminus \bigcup A_i$ exists and $(\{w\}, \varnothing)$ strictly extends the family [5]. The 40/8 census counts for $W = \{2, 3, 4\}$ stay [COMPUTED]×2 (cited, not formalized — no half-formalization) [5, 9].

5 Discussion

Significance. The void-frame theory is the program's forcing framework stripped to its algebra: frames fire or they do not; the void frames are the identity; the defect datum measures exactly how far a product is from being a homomorphism; and H1 says no nonzero additive measurement of the defect monoid exists at all. The surprise is not the rigidity (it is classical idempotent-collapse) but that the parameterized calculus closes exactly: 3^n, 5^n, $2^n \cdot \mathrm{Bell}(n)$ with zero violations across two independent code paths [7]. The C1/C2 result is the honest counterpoint: maximality does not imply splitting — a distributional fact (Figure 2) the Lean witness pins exactly.

Novelty. All objects are program-invented (void frames, (T, F, C), N', d); novelty is narrow: first statements about these objects. Classical carve-outs are stated: M2/H1 idempotent-collapse [4, 11]; semilattice = idempotent commutative monoid [5]; $2^n \cdot \mathrm{Bell}(n)$ = OEIS A055882 (recorded, not claimed) [7]; Bell/Stirling numbers. No blanket algebraic novelty is claimed. Web-checks (2026-10-11) found no "void frame" / "defect-additive monoid" / "void monoid" in the literature (hits are unrelated: programming monoids, valuation-theory defect, intersecting families) [4, 5, 7].

Limitations. (i) The 217/246 void census and all pair censuses are [COMPUTED]×2, not Lean-formalized: the machine checks the theorems about the datum, not the census itself (no half-formalization) [8, 9]. (ii) Q-34's scoped verdict explicitly withholds: the 1,097-quad cell-0 census as a Lean claim, code independence, "every $k \equiv 0 \bmod 35$", and full R7 acceptance — these are UNVERIFIED/OPEN, not proved here [8]. (iii) The `VoidMonoidParam` theorems are review-staged (build-clean, gate-passed, Q-42/Q-45 SURVIVES, promotion pending); T3/T4/H3 formalization is owed to a dedicated lane [8, 9]. (iv) Beyond the exclusion lemma, void-ness is contingent edge data ([HEURISTIC]-leaning); no further structural restriction was found [2]. (v) The X-side enlargement ($X \neq A^\star$, τ not forced) is a different object class, unexamined [OPEN] [2].

6 Conclusion

Void frames, abundant in the swept universe (217/246), are the union identity of the frame algebra and exactly the zero triple of the (T, F, C) datum (V1–V8) [PROVED]. The void-frame monoid N'_W carries an exact defect calculus: the defect formula (6), parameterized closed forms (P1–P2) [PROVED*]-staged, and the product defect-split with its honest overlap correction (T1/T1′/T2) [PROVED*]-staged. The defect homomorphism theory is rigid: the only homomorphism to a cancellative target is zero (H1) [PROVED]; literal products of nontrivial families are never defect-additive (C4) [PROVED]; and maximal families need not split — the mixed ones are counted exactly (C1/C2) [PROVED*/COMPUTED]. The pairwise-disjoint repair theorem and the maximal-family tiling theorem are [PROVED]. Everything machine-checked is formalized in Lean 4.34.1 + Mathlib v4.34.1 with zero sorrys and standard axioms only; every census is cited [COMPUTED] data, never called a proof. The natural next steps are the owed formalizations (P2(iii)(b), P2(iv), T3/T4/H3), the X-side void theory, and the $k \equiv 0 \bmod 35$ general obstruction.

7 References

References

[1] D. Blair, *Unsolved Math — Round 280: abstract lock theory [PROVED*]*, program dossier, 2026-10-11. (Supplied datum: abstract lock frame, Theorems 1–4, 472-frame census.)

[2] D. Blair, *Unsolved Math — Round 283: void locks — the abstract prediction realized; void-exclusion lemma [PROVED*]; void census [COMPUTED]*×2, program dossier, 2026-10-10. (Supplied datum: void-search universe, 217/246 void census, pattern census, Void-Exclusion Lemma, representative void frames.)

[3] D. Blair, *Unsolved Math — Round 289: void frames under union forcing (R283 × R285 cross) + the cell-0 gap CLOSED*, program dossier, 2026-10-11. (Supplied datum: $(T, F, C) = (|u|, 0, |u|)$ datum, V1–V8, 30,381-pair census, $k_0 = 0$ impossibility, flagship $(5, 553, 70, 2, 910, 2765)$ cell-0 family, 1,097-quad census, Q-34 scope notes.)

[4] D. Blair, *Unsolved Math — Round 291: the void-frame algebra (void monoid)*, program dossier, 2026-10-11. (Supplied datum: N', ι, d, π_1/π_2, S1/M1/M2/M3/X1/X2/F1, E/C/V classes, 60,516-pair census.)

[5] D. Blair, *Unsolved Math — Round 296: defect core of the void-frame monoid*, program dossier, 2026-10-11. (Supplied datum: D1/D2/D3, 27-submonoid classification, 40-family census, draft-error log.)

[6] D. Blair, *Unsolved Math — Round 299: D2(c) wording repair (Q-39)*, program dossier, 2026-10-11. (Supplied datum: verbatim Q-39 repair, 88-family re-verification.)

[7] D. Blair, *Unsolved Math — Round 300: parameterized defect calculus*, program dossier, 2026-10-11. (Supplied datum: N'_W parameterized, P1, P2(i)–(iv), closed forms, $n = 0 \ldots 4$ census, Q-42 gate.)

[8] D. Blair, *Unsolved Math — Round 314: Lean + rule-9 lane*, program dossier, 2026-10-11. (Supplied datum: `VoidMonoidParam.lean` fold record, Q-42/Q-45 SURVIVES, R314 build gate, Q-34 scope notes, owed-formalization list.)

[9] D. Blair, *Unsolved Math — Round 318: Lean + rule-9 lane*, program dossier, 2026-10-11. (Supplied datum: `VoidMonoid.lean` fold record (`d2c_not_closed`, `d2c_generated_loses_disjoint`, `maximal_family_tiles`), R318 build gate, owed-T3/T4/H3 note.)

[10] D. Blair, *Unsolved Math — Round 319: R303 product-defect theory filed + maximal splitting classification*, program dossier, 2026-10-11. (Supplied datum: T1/T1′/T2 hand proofs, H1 hand proof, C1 split boundary, C2 mixed counts $(1,1)\colon 4$, $(2,1)\colon 24$, $(2,2)\colon 176$, C3 record correction, C4 literal-product theorem.)

[11] D. Blair, *Unsolved Math — Round 322: Lean + rule-9 lane*, program dossier, 2026-10-11. (Supplied datum: `VoidDefectHom.lean` fold record (`defect_hom_rigidity`, `literal_product_never`, `split_boundary_witness`, `mixed_count`), Q-58 SURVIVES, 34-theorem batch gate, axiom audit.)

[12] D. Blair, *Unsolved Math — Round 310: Lean + rule-9 lane*, program dossier, 2026-10-11. (Supplied datum: bakkie full-library build gate, exit 0, 17,936 jobs, axiom audit, [PROVED] promotions of VoidLock/VoidUnion.)